

SELECT PUBLICATIONS

109 refereed and non-refereed publications, 13,090 citations, h-index 40 (INSPIRE-HEP, July 2026)

Recent Refereed Publications (2024-2026)

- S. Abubakar et al. (NOvA Collaboration), “Measurement of π^0 Production in $\bar{\nu}_\mu$ Charged-Current Interactions in the NOvA Near Detector”, Phys. Rev. D 113, 112008 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Ionization-based search for magnetic monopoles using the NOvA Far Detector”, Phys. Rev. D 113, 112014 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Identification of low-energy kaons in the ProtoDUNE-SP detector”, Phys. Rev. D 113, 052004 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Operation of a Modular 3D-Pixelated Liquid Argon Time-Projection Chamber in a Neutrino Beam”, Instruments 10, 18 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Precision Measurement of Neutrino Oscillation Parameters with 10 Years of Data from the NOvA Experiment”, Phys. Rev. Lett. 136 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Explanation of the seasonal variation of cosmic multiple muon events observed with the NOvA Near Detector”, Phys. Rev. D 113 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Measurement of $d^2\sigma/d|q|dE_{avail}$ in charged current neutrino-nucleus interactions at $\langle E_\nu \rangle = 1.86$ GeV using the NOvA Near Detector”, Phys. Rev. D 111, 052009 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations in NOvA”, Phys. Rev. Lett. 134, 081804 (2025)
- M. A. Acero et al. (NOvA and T2K Collaborations), “Joint neutrino oscillation analysis from the T2K and NOvA experiments”, Nature 646, 818 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Monte Carlo method for constructing confidence intervals with unconstrained and constrained nuisance parameters in the NOvA experiment”, JINST 20 02, T02001 (2025)
- S. Abbaslu et al. (DUNE Collaboration), “Spatial and temporal evaluations of the liquid argon purity in ProtoDUNE-SP”, JINST 20 09, P09008 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The track-length extension fitting algorithm for energy measurement of interacting particles in liquid argon TPCs and its performance with ProtoDUNE-SP data”, JINST 20 02, P02021 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “Supernova pointing capabilities of DUNE”, Phys. Rev. D 111 9, 092006 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Search for CP-violating Neutrino Non-Standard Interactions with the NOvA Experiment”, Phys. Rev. Lett. 133 20, 201802 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “First Measurement of the Total Inelastic Cross-Section of Positively-Charged Kaons on Argon at Energies Between 5.0 and 7.5 GeV”, Phys. Rev. D 110 9, 092011 (2024)
- M. A. Acero et al. (NOvA Collaboration), “Expanding neutrino oscillation parameter measurements in NOvA using a Bayesian approach”, Phys. Rev. D 110, 012005 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Far Detector Vertical Drift Technology. Technical Design Report”, JINST 19 08, T08004 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “DUNE Phase II: scientific opportunities, detector concepts, technological solutions”, JINST 19 12, P12005 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “Performance of a Modular Ton-Scale Pixel-Readout Liquid Argon Time Projection Chamber”, Instruments 8 3, 41 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “Doping liquid argon with xenon in ProtoDUNE Single-Phase: effects on scintillation light”, JINST 19 08, P08005 (2024)

Publications (2020-2023)

- M. A. Acero et al. (NOvA Collaboration), “Measurement of the ν_e -Nucleus Charged-Current Double-Differential Cross Section at $\langle E_\nu \rangle = 2.4$ GeV using NOvA”, Phys. Rev. Lett. 130 (2023) 5, 051802 (2023)

- A. Abed Abud et al. (DUNE Collaboration), “Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino Experiment”, *Phys. Rev. D* 107, 112012 (2023)
- M. A. Acero et al. (NOvA Collaboration), “Improved measurement of neutrino oscillation parameters by the NOvA experiment”, *Phys. Rev. D* 106, 032004 (2022)
- A. Abed Abud et al. (DUNE Collaboration), “Design, construction and operation of the ProtoDUNE-SP Liquid Argon TPC”, *JINST* 17 01, P01005 (2022)
- M. A. Acero et al. (NOvA Collaboration), “Search for Active-Sterile Antineutrino Mixing Using Neutral-Current Interactions with the NOvA Experiment”, *Phys. Rev. Lett.* 127, 201801 (2021)
- B. Abi et al. (DUNE Collaboration), “Supernova neutrino burst detection with the Deep Underground Neutrino Experiment”, *Eur. Phys. J. C* 81 5, 423 (2021)
- B. Abi et al. (DUNE Collaboration), “First results on ProtoDUNE-SP liquid argon time projection chamber performance from a beam test at the CERN Neutrino Platform”, *JINST* 15 12, P12004 (2020)
- B. Abi et al. (DUNE Collaboration), “Long-baseline neutrino oscillation physics potential of the DUNE experiment”, *Eur. Phys. J. C* 80 10, 978 (2020)
- B. Abi et al. (DUNE Collaboration), “Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume IV: Far Detector Single-phase Technology”, *JINST* 15 08, T08010 (2020)

Publications (2016-2019)

- M. A. Acero et al. (NOvA Collaboration), “First Measurement of Neutrino Oscillation Parameters using Neutrinos and Antineutrinos by NOvA”, *Phys. Rev. Lett.* 123 151803 (2019)
- M. A. Acero et al. (NOvA Collaboration), “New constraints on oscillation parameters from ν_e appearance and ν_μ disappearance in the NOvA experiment”, *Phys. Rev. D* 98, 032012 (2018)
- P. Adamson et al. (NOvA Collaboration), “Search for active-sterile neutrino mixing using neutral-current interactions in NOvA”, *Phys. Rev. D* 96, 072006 (2017)
- P. Adamson et al. (NOvA Collaboration), “Constraints on Oscillation Parameters from ν_e Appearance and ν_μ Disappearance in NOvA”, *Phys. Rev. Lett.* 118, 231801 (2017)
- P. Adamson et al. (NOvA Collaboration), “Measurement of the neutrino mixing angle θ_{23} in NOvA”, *Phys. Rev. Lett.* 118, 151802 (2017)
- P. Adamson et al. (NOvA Collaboration), “First Measurement of Electron Neutrino Appearance in NOvA”, *Phys. Rev. Lett.* 116, 151806 (2016)
- P. Adamson et al. (NOvA Collaboration), “First Measurement of muon-neutrino disappearance in NOvA”, *Phys. Rev. D* 93, 051104(R) (2016)

T2K Collaboration

- D. Allan et al. (T2K Collaboration), “The Electromagnetic Calorimeter for the T2K Near Detector”, *JINST* 8 P10019 (2013)
- K. Abe et al. (T2K Collaboration), “First Muon-Neutrino Disappearance Study with an Off-Axis Beam”, *Phys. Rev. D* 85, 031103 (2012)
- K. Abe et al. (T2K Collaboration), “Indication of Electron Neutrino Appearance from an Accelerator-produced Off-axis Muon Neutrino Beam”, *Phys. Rev. Lett.* 107, 041801 (2011)

Selected Non-refereed Publications

- S. Abbaslu et al. (DUNE Collaboration), “Operation and performance of ProtoDUNE Dual Phase liquid argon time projection chamber”, *arXiv:2607.15927* (2026)
- S. Abubakar et al. (NOvA Collaboration), “Determining Neutrino Mass Ordering with NOvA and Upcoming JUNO Measurements”, *arXiv:2606.14121* (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Design of the DUNE horizontal drift far detector charge readout electronics and performance in ProtoDUNE-HD”, *arXiv:2604.23966* (2026)
- S. Abubakar et al. (NOvA Collaboration), “First inclusive triple-differential measurement of the muon-antineutrino charged-current cross section using the NOvA Near Detector”, *arXiv:2603.06718* (2026)

- T. K. Aarrestad et al. (HEP Community), “Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision”, arXiv:2602.17582 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Reconstruction of atmospheric neutrinos in DUNE’s horizontal-drift far-detector module”, arXiv:2601.05697 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Measurement of Exclusive π^+ -argon Interactions Using ProtoDUNE-SP”, arXiv:2511.13462 (2025)
- S. Abbaslu et al. (DUNE Collaboration), “First Measurement of π^+ -Ar and p -Ar Total Inelastic Cross Sections in the Sub-GeV Energy Regime with ProtoDUNE-SP Data”, arXiv:2511.11925 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “European Contributions to Fermilab Accelerator Upgrades and Facilities for the DUNE Experiment”, arXiv:2503.23744 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “DUNE Software and Computing Research and Development”, arXiv:2503.23743 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Science Program”, arXiv:2503.23291 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Phase II Detectors”, arXiv:2503.23293 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Measurement of the double-differential cross section of muon-neutrino charged-current interactions with low hadronic energy in the NOvA Near Detector”, arXiv:2410.10222 (2024)
- Abed Abud et al. (DUNE Collaboration), “DUNE Offline Computing Conceptual Design Report”, arXiv:2210.15665 (2022)
- V. Daniel Elvira et al., “The Future of High Energy Physics Software and Computing”, Snowmass 2021, arXiv:2210.05822 (2022)
- G.S. Davies, P. Onyisi, and A. Roberts, “CompF5: End User Analysis Topical Group Report”, Snowmass 2021, arXiv:2209.14984 (2022)
- A. Abed Abud et al. (DUNE Collaboration), “Snowmass Neutrino Frontier: DUNE Physics Summary”, Snowmass 2021, arXiv:2203.06100 (2022)
- B. Abi et al. (DUNE Collaboration), “Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume II: DUNE Physics”, arXiv:2002.03005 (2020)

Full INSPIRE Publication List

- Operation and performance of ProtoDUNE Dual Phase liquid argon time projection chamber, arXiv:2607.15927 (2026)
- Determining Neutrino Mass Ordering with NOvA and Upcoming JUNO Measurements, arXiv:2606.14121 (2026)
- Design of the DUNE horizontal drift far detector charge readout electronics and performance in ProtoDUNE-HD, arXiv:2604.23966 (2026)
- First inclusive triple-differential measurement of the muon-antineutrino charged-current cross section using the NOvA Near Detector, arXiv:2603.06718 (2026)
- Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision, arXiv:2602.17582 (2026)
- Reconstruction of atmospheric neutrinos in DUNE's horizontal-drift far-detector module, arXiv:2601.05697 (2026)
- Ionization-based search for magnetic monopoles using the NOvA Far Detector, Phys. Rev. D 113, 112014 (2026), arXiv:2512.20294
- Measurement of Exclusive + -argon Interactions Using ProtoDUNE-SP, arXiv:2511.13462 (2025)
- First Measurement of + -Ar and p -Ar Total Inelastic Cross Sections in the Sub-GeV Energy Regime with ProtoDUNE-SP Data, arXiv:2511.11925 (2025)
- Measurement of 0 Production in bar Charged-Current Interactions in the NOvA Near Detector, arXiv:2511.05807 (2025)
- Joint neutrino oscillation analysis from the T2K and NOvA experiments, arXiv:2510.19888 (2025)
- Identification of low-energy kaons in the ProtoDUNE-SP detector, arXiv:2510.08380 (2026)
- Operation of a Modular 3D-Pixelated Liquid Argon Time-Projection Chamber in a Neutrino Beam, arXiv:2509.07012 (2026)
- Precision Measurement of Neutrino Oscillation Parameters with 10 Years of Data from the NOvA Experiment, arXiv:2509.04361 (2026)
- Explanation of the seasonal variation of cosmic multiple muon events observed with the NOvA Near Detector, arXiv:2508.04434 (2026)
- Spatial and temporal evaluations of the liquid argon purity in ProtoDUNE-SP, arXiv:2507.08586 (2025)
- European Contributions to Fermilab Accelerator Upgrades and Facilities for the DUNE Experiment, arXiv:2503.23744 (2025)
- DUNE Software and Computing Research and Development, arXiv:2503.23743 (2025)
- The DUNE Science Program, arXiv:2503.23291 (2025)
- The DUNE Phase II Detectors, arXiv:2503.23293 (2025)
- Measurement of the double-differential cross section of muon-neutrino charged-current interactions with low hadronic energy in the NOvA Near Detector, arXiv:2410.10222 (2024)
- Measurement of $d^2/d^2q^2 dE_{\text{avail}}$ in charged current -nucleus interactions at $E = 1.86$ GeV using the NOvA Near Detector, arXiv:2410.05526 (2025)
- The track-length extension fitting algorithm for energy measurement of interacting particles in liquid argon TPCs and its performance with ProtoDUNE-SP data, arXiv:2409.18288 (2025)
- Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations in NOvA, arXiv:2409.04553 (2025)
- DUNE Phase II: scientific opportunities, detector concepts, technological solutions, arXiv:2408.12725 (2024)
- First measurement of the total inelastic cross section of positively charged kaons on argon at energies between 5.0 and 7.5 GeV, arXiv:2408.00582 (2024)
- Supernova pointing capabilities of DUNE, arXiv:2407.10339 (2025)
- Search for CP-Violating Neutrino Nonstandard Interactions with the NOvA Experiment, arXiv:2403.07266 (2024)
- Performance of a Modular Ton-Scale Pixel-Readout Liquid Argon Time Projection Chamber, arXiv:2403.03212 (2024)
- Doping liquid argon with xenon in ProtoDUNE Single-Phase: effects on scintillation light, arXiv:2402.01568 (2024)
- The DUNE Far Detector Vertical Drift Technology. Technical Design Report, arXiv:2312.03130

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- Expanding neutrino oscillation parameter measurements in NOvA using a Bayesian approach, arXiv:2311.07835 (2024)
 - Measurement of charged-current inclusive $\bar{\nu}_e$ production in the NOvA near detector, arXiv:2306.04028 (2023)
 - Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino Experiment, arXiv:2303.17007 (2023)
 - Highly-parallelized simulation of a pixelated LArTPC on a GPU, arXiv:2212.09807 (2023)
 - Identification and reconstruction of low-energy electrons in the ProtoDUNE-SP detector, arXiv:2211.01166 (2023)
 - DUNE Offline Computing Conceptual Design Report, arXiv:2210.15665 (2022)
 - CompF5: End User Analysis Topical Group Report, arXiv:2209.14984 (2022)
 - Monte Carlo method for constructing confidence intervals with unconstrained and constrained nuisance parameters in the NOvA experiment, arXiv:2207.14353 (2025)
 - Reconstruction of interactions in the ProtoDUNE-SP detector with Pandora, arXiv:2206.14521 (2023)
 - Measurement of the ν_e -Nucleus Charged-Current Double-Differential Cross Section at $\sqrt{s} = 2.4$ GeV using NOvA, arXiv:2206.10585 (2023)
 - Separation of track- and shower-like energy deposits in ProtoDUNE-SP using a convolutional neural network, arXiv:2203.17053 (2022)
 - Scintillation light detection in the 6-m drift-length ProtoDUNE Dual Phase liquid argon TPC, arXiv:2203.16134 (2022)
 - Dark-matter And Neutrino Computation Explored (DANCE) Community Input to Snowmass, arXiv:2203.08338 (2022)
 - White paper on light sterile neutrino searches and related phenomenology, arXiv:2203.07323 (2024)
 - Snowmass Neutrino Frontier: DUNE Physics Summary, arXiv:2203.06100 (2022)
 - A Gaseous Argon-Based Near Detector to Enhance the Physics Capabilities of DUNE, arXiv:2203.06281 (2022)
 - Measurement of the double-differential muon-neutrino charged-current inclusive cross section in the NOvA near detector, arXiv:2109.12220 (2023)
 - Design, construction and operation of the ProtoDUNE-SP Liquid Argon TPC, arXiv:2108.01902 (2022)
 - Searching for solar KDAR with DUNE, arXiv:2107.09109 (2021)
 - Extended search for supernovalike neutrinos in NOvA coincident with LIGO/Virgo detections, arXiv:2106.06035 (2021)
 - Search for Active-Sterile Antineutrino Mixing Using Neutral-Current Interactions with the NOvA Experiment, arXiv:2106.04673 (2021)
 - Seasonal variation of multiple-muon cosmic ray air showers observed in the NOvA detector on the surface, arXiv:2105.03848 (2021)
 - Deep Underground Neutrino Experiment (DUNE) Near Detector Conceptual Design Report, arXiv:2103.13910 (2021)
 - Experiment Simulation Configurations Approximating DUNE TDR, arXiv:2103.04797 (2021)
 - Search for slow magnetic monopoles with the NOvA detector on the surface, arXiv:2009.04867 (2021)
 - Prospects for beyond the Standard Model physics searches at the Deep Underground Neutrino Experiment, arXiv:2008.12769 (2021)
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 - First results on ProtoDUNE-SP liquid argon time projection chamber performance from a beam test at the CERN Neutrino Platform, arXiv:2007.06722 (2020)
 - Neutrino interaction classification with a convolutional neural network in the DUNE far detector, arXiv:2006.15052 (2020)
 - Long-baseline neutrino oscillation physics potential of the DUNE experiment, arXiv:2006.16043

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- Adjusting neutrino interaction models and evaluating uncertainties using NOvA near detector data, arXiv:2006.08727 (2020)
- Supernova neutrino detection in NOvA, arXiv:2005.07155 (2020)
- Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume III: DUNE Far Detector Technical Coordination, arXiv:2002.03008 (2020)
- Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume I Introduction to DUNE, arXiv:2002.02967 (2020)
- Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume II: DUNE Physics, arXiv:2002.03005 (2020)
- Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume IV: Far Detector Single-phase Technology, arXiv:2002.03010 (2020)
- Search for multimessenger signals in NOvA coincident with LIGO/Virgo detections, arXiv:2001.07240 (2020)
- First Measurement of Neutrino Oscillation Parameters using Neutrinos and Antineutrinos by NOvA, arXiv:1906.04907 (2019)
- Observation of seasonal variation of atmospheric multiple-muon events in the NOvA Near Detector, arXiv:1904.12975 (2019)
- Measurement of neutrino-induced neutral-current coherent π^0 production in the NOvA near detector, arXiv:1902.00558 (2020)
- The DUNE Far Detector Interim Design Report Volume 1: Physics, Technology and Strategies, arXiv:1807.10334 (2018)
- The DUNE Far Detector Interim Design Report, Volume 2: Single-Phase Module, arXiv:1807.10327 (2018)
- The DUNE Far Detector Interim Design Report, Volume 3: Dual-Phase Module, arXiv:1807.10340 (2018)
- New constraints on oscillation parameters from ν_e appearance and ν_μ disappearance in the NOvA experiment, arXiv:1806.00096 (2018)
- Results and Prospects from NOvA (2018)
- Event Reconstruction in the NOvA Experiment, arXiv:1710.03772 (2017)
- Sterile Neutrino Search in the NOvA Far Detector, arXiv:1710.01280 (2017)
- NOvA Short-Baseline Tau Neutrino Appearance Search, arXiv:1710.00295 (2017)
- A search for sterile neutrinos at the NOvA Far Detector (2017)
- The Single-Phase ProtoDUNE Technical Design Report, arXiv:1706.07081 (2017)
- Search for active-sterile neutrino mixing using neutral-current interactions in NOvA, arXiv:1706.04592 (2017)
- Constraints on Oscillation Parameters from ν_e Appearance and ν_μ Disappearance in NOvA, arXiv:1703.03328 (2017)
- Measurement of the neutrino mixing angle θ_{23} in NOvA, arXiv:1701.05891 (2017)
- Searches for Sterile Neutrinos with NOvA (2016)
- Physics potential of a long-baseline neutrino oscillation experiment using a J-PARC neutrino beam and Hyper-Kamiokande, arXiv:1502.05199 (2015)
- A Long Baseline Neutrino Oscillation Experiment Using J-PARC Neutrino Beam and Hyper-Kamiokande, arXiv:1412.4673 (2014)
- Expression of Interest: The Atmospheric Neutrino Neutron Interaction Experiment (ANNIE), arXiv:1402.6411 (2014)
- Hyper-Kamiokande Physics Opportunities, arXiv:1309.0184 (2013)
- The Electromagnetic Calorimeter for the T2K Near Detector ND280, arXiv:1308.3445 (2013)
- The Long-Baseline Neutrino Experiment: Exploring Fundamental Symmetries of the Universe, arXiv:1307.7335 (2013)
- Scientific Opportunities with the Long-Baseline Neutrino Experiment (2013)
- CHerenkov detectors In mine PitS (CHIPS) Letter of Intent to FNAL, arXiv:1307.5918 (2013)
- Cosmic Ray Backgrounds in an LBNE far detector on the Surface, arXiv:1206.3561 (2012)
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- The Long Baseline Neutrino Experiment (LBNE) Water Cherenkov Detector (WCD) Conceptual Design Report (CDR), arXiv:1204.2295 (2012)
- First Muon-Neutrino Disappearance Study with an Off-Axis Beam, arXiv:1201.1386 (2012)
- Understanding and Calibrating the Calorimeter for the T2K Near Detector (2012)
- Measurements of the T2K neutrino beam properties using the INGRID on-axis near detector, arXiv:1111.3119 (2012)
- NOvA: Present and Future, arXiv:1110.0112 (2011)
- The 2010 Interim Report of the Long-Baseline Neutrino Experiment Collaboration Physics Working Groups, arXiv:1110.6249 (2011)
- The T2K Experiment, arXiv:1106.1238 (2011)
- Indication of Electron Neutrino Appearance from an Accelerator-produced Off-axis Muon Neutrino Beam, arXiv:1106.2822 (2011)
- Understanding and calibrating the DS-ECal for T2K's ND280 (2011)
- The T2K experiment and downstream electromagnetic calorimeter (2008)
- Radiation hardness studies in a CCD with high-speed column parallel readout, arXiv:0711.3561 (2008)
- Simulations of the Temperature Dependence of the Charge Transfer Inefficiency in a High-Speed CCD (2007)
- Progress with vertex detector sensors for the International Linear Collider (2007)
- Radiation Hardness of CCD Vertex Detectors for the ILC, arXiv:physics/0611272 (2007)